

[41/A-15]

SEAT No. _____

SARDAR PATEL UNIVERSITY

External Examination (CBCS)

M. Sc. - Information Technology (Integrated)

IInd - Semester (CBCS)

PS02CIIT03: Advanced 'C' and Introduction to Data Structures

18th April, Tuesday - 2017

Time : 10:00 am to 1:00 pm

Total Marks :70

Q-1 Select an appropriate option.

10

1. Which operator is used with a pointer to access the value of the variable whose address is contained in the pointer?
(a) Address (&) (b) Indirection (*) (c) Assignment (=) (d) Selection (->)
2. Which of the following defines a pointer variable to an integer?
(a) int &ptr (b) int **ptr (c) int &&ptr (d) int *ptr
3. Which of the following is not a derived data type?
(a) Arrays (b) Pointers (c) Float (d) Structure
4. Which of the following allows a portion of memory to be shared by different types of data?
(a) Array (b) Union (c) Structure (d) File
5. Which one of the following is valid for opening a file for only reading?
(a) fopen (filenm, "r"); (b) fopen (filenm, "r");
(c) fopen (filenm, "ra"); (d) fopen (filenm, "read");
6. Files are a _____ type of Data Structure.
(a) Linear (b) Non-Primitive (c) Primitive (d) Non-linear
7. A stack is _____ type of data structure.
(a) Linear (b) Non-linear (c) Both (a) and (b) (d) None of these
8. A linked list is _____ type of data structure.
(a) Linear (b) Non-linear (c) Both (a) and (b) (d) None of these
9. A data structure in which insertion and deletion of an elements occurs at both the end is known as _____.
(a) Stack (b) Queue (c) Priority Queue (d) Deque
10. A linked list in which last node pointing to the first node is known as _____.
(a) Singly linked list (b) Doubly linked list
(c) Circular linked list (d) None of these

Q-2 Answer the following questions. (Attempt any TEN)

20

1. Define: indirection operator, pointer variable
2. Differentiate: structure and union

3. List out benefits of pointers.
4. List file modes available to manage the file in C.
5. Explain typedef in brief with suitable example.
6. Explain the fclose() function with example.
7. What is non-primitive Data Structure?
8. Draw the Hierarchical Structure of Data Structure.
9. State various applications of Stack.
10. State various types of queue.
11. What is a singly linked list?
12. Differentiate between stack and queue data structure.

Q-3

- (a) What are pointers? How can they be used with arrays? Explain pointer to an array using appropriate examples. 5
- (b) Write a short note on dynamic memory allocation. 5

OR

Q-3

- (a) What is structure? Explain its definition, declaration and assigning values to members of structure. 5
- (b) Explain pointer to structure array using appropriate example. 5

Q-4

- (a) What is union? Explain its definition, declaration and assigning values to members of union. 5
- (b) Explain fprintf and fscanf function with example. 5

OR

Q-4

- (a) Explain storage representation of union. How a member of union is assigned an initial value? Explain. 5
- (b) Explain the getw and putw function with example. 5

Q-5

- (a) What is data structure? Also write advantages of data structure. 5
- (b) Write an algorithm: (i) PUSH operation (ii) PEEP operation. 5

OR

Q-5

- (a) Write a short note on primitive data structure operations. 5
- (b) Write an algorithm: (i) POP operation (ii) CHANGE operation 5

- Q-6 Write an algorithm for the followings: 10
- (i) Insert an element at the beginning of a singly linked list.
 - (ii) Insert an element at the end of a singly linked list.

OR

- Q-6 Write an algorithm for the followings: 10
- (i) Insert an element into a simple queue.
 - (ii) Delete an element from a simple queue.

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